Handy Helper Umbrella

8-Panel Field Reference for HVAC & Electrical Trades

1 Safety First

LOTO, PPE & Wet-Work Precautions

Lockout/Tagout (LOTO)[1]

1. Notify anyone affected
2. Shut down by normal means
3. Isolate — disconnect/break energy source
4. Lock out — apply individual lock + tag
5. Verify de-energized — test meter on known source first, then circuit
6. Release — only the person who applied the lock removes it

Wet-Work Precautions[1]

Never pressure wash energized electrical components. Cover motors, contactors, control boards. Keep water away from disconnects/panels. Allow equipment to dry before re-energizing. Use GFCI-protected power.

PPE Checklist

Safety glasses/face shield | Electrical-rated gloves (Class 0) | Hearing protection | EH-rated boots | Long sleeves for brazing

1. OSHA Control of Hazardous Energy: https://www.osha.gov/control-hazardous-energy

2 Refrigeration Cycle

The 4 Stages of the Cycle

The 4 Stages[2,3]

1. Compressor — raises pressure & temperature of vapor
2. Condenser — rejects heat, vapor becomes liquid
3. Metering Device — drops pressure, liquid becomes mixed phase
4. Evaporator — absorbs heat, refrigerant becomes vapor

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COMPRESSOR
^ v
Low-P High-P
vapor vapor
^ v
EVAP CONDENSER
^ v
<-- METERING DEVICE -->
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R410A PT Conversions[10]

350-450 PSIG corresponds to 106.8-125.4°F saturated
100-140 PSIG corresponds to 31.4-48.9°F saturated
Conversions only, not operating or charging targets.

Metering Devices: TXV (best efficiency) | Fixed piston (charge-sensitive) | Cap tube (small systems) | EEV (electronic, variable)

2. HVAC School: https://www.hvacrschool.com/
3. Anthony's Cool School HVAC Training: https://hvac.anthonyscoolschool.com/
10. Honeywell AZ-20 PT chart (Feb 2019): https://prod-edam.honeywell.com/content/dam/honeywell-edam/pmt/oneam/en-us/refrigerants/documents/pmt-am-genetron-aZ20-pressure-temp-chart-tech-tool1.pdf

3 Superheat & Subcooling

Charge & Performance Diagnostics

Superheat = Suction Line Temp – Sat. Suction Temp[4]
Measures evaporator performance

Superheat	Likely Cause
High (>15-20F)	Undercharge, restriction, low airflow
Low (<5-10F)	Overcharge, overfeeding, high airflow
Normal	Target varies by conditions and manufacturer chart

Subcooling = Sat. Liquid Temp – Liquid Line Temp[4]
Measures condenser performance

Subcooling	Likely Cause
High (>15-20F)	Overcharge, liquid line restriction
Low (<5-10F)	Undercharge, condenser airflow issue
Normal (8-12F)	Verify with manufacturer specs

Steps: Measure line temps | Read gauges | PT chart for sat temps | Calculate | Compare to mfr specs

4. HVAC School - Superheat & Subcooling: https://www.hvacrschool.com/

4 Thermostat Wiring

Low-Voltage Terminal Reference

Terminal Reference[5]

Term	Color	Function
R/RH/RC	Red	24V power (heat/cool)
C	Blue	Common return
Y/Y1	Yellow	Cooling -> contactor
G	Green	Fan / blower
W/W1	White	Heating
O	Orange	Rev. valve - cool (most)
B	Dk Blue	Rev. valve - heat (Ruud/Rheem)
W2	White	Aux / 2nd-stage heat

Common Failures

No 24V at stat: Check transformer, fuse, float switch
Stat blank: Check R-C for 24V, C wire connection
Fan won't stop: G wire shorted, relay stuck
No cooling: Y not reaching contactor, float/pressure switch
Rev. valve wrong: Check O/B — energize-in-cool vs heat

Float switch: Breaking R kills stat + defrost board. Breaking Y kills cooling only.

5. HVAC School - Low Voltage Wiring: https://www.hvacrschool.com/videos/understanding-low-voltage-wiring-for-ac-heat-pumps-3d/

5 Contactor & Capacitor

Motor Circuit Testing

Contactor[6]

Coil: 24V from stat Y -> safety switches -> contactor coil
Contacts: 240V through contacts to compressor + condenser fan
Test: Click sound, 24V across coil, ~0V across closed contacts

Capacitor Terminals

Terminal	Connects To
C (Common)	Run winding + power
HERM	Compressor
FAN	Condenser fan motor

WARNING: Capacitors hold charge even when power is off!
Discharge with 20k-ohm resistor across terminals. Wait 30s, short again. Never use a screwdriver.

Testing

Visual: Bulging/leaking = replace | MFD: +/-10% of rating | Ground: Terminal to case = OL (infinite)

6. HVAC School: https://www.hvacrschool.com/

6 Heat Pump & Defrost

Sequence & Troubleshooting

Heating Mode Sequence[7]

1. Stat calls heat (W/Y energizes)
2. Reversing valve positions for heating (de-energized on most brands)
3. Compressor starts — outdoor coil = evaporator (absorbs heat)
4. Indoor coil = condenser (rejects heat)
5. Supplemental heat (W2) engages if needed

Defrost Cycle

Triggers when ice builds on outdoor coil (typically <40F outdoor). Defrost board reverses to cooling mode, outdoor fan stops, heat strips energize. Duration: 5-15 min.

Problem	Check
Never defrosts	Defrost sensor/board, reversing valve
Defrosts too often	Sensor location/accuracy
Won't exit defrost	Sensor stuck, board relay
Ice on coil	Low charge, low airflow, defrost fail

Rev. valve: Most brands energize O in cooling. Ruud/Rheem energize B in heating.

7. HVAC School: https://www.hvacrschool.com/
Anthony's Cool School: https://hvac.anthonyscoolschool.com/

7 Electrical Formulas

Ohm's Law, Power & Voltage Drop

Ohm's Law[8]

V = I x R I = V / R R = V / I

Power (Single Phase)

P = V x I P = I² x R P = V² / R

Three-Phase Power

P = V x I x 1.732 x PF (PF typically 0.8-0.95)

Voltage Drop (Copper)[8]

VD = 2 x 12.9 x I x D / CM

K=12.9 (copper) | I=amps | D=one-way feet | CM=circular mils

AWG	Ampacity (60C)	CM
14	15A	4,107
12	20A	6,530
10	30A	10,380
8	40A	16,510
6	55A	26,240

Motor FLA estimate: 240V 1-phase ~= HP x 4.5 | 3-phase ~= HP x 2.5

Always verify against NEC Article 310 and local code.[8]

8. NFPA 70 - National Electrical Code: https://www.nfpa.org/codes-and-standards/detail?code=70

8 Troubleshooting Flow

Decision Trees & PM Checklist

No Cooling[9]

Stat on COOL, below room temp?

+-- No -> Check stat, batteries, display

+-- Yes -> Outdoor unit running?

+-- No -> Breaker -> disconnect -> contactor -> cap -> compressor

+-- Yes -> Check suction/head pressure

+-- Low S, low H -> Low charge / leak

+-- High S, high H -> Restricted airflow / overcharge

+-- Low S, high H -> Restriction (filter drier, TXV)

+-- High S, low H -> Bad compressor valves

Short Cycling

Check: stat location (drafts), filter, coil freeze-up, low charge, oversized equip, pressure switch

Pressure Washing PM[1]

Cover electricals | Use low pressure, wide-fan nozzle | Prefer garden-hose pressure when possible | Manufacturer-approved coil cleaner per label | Top-down for condenser | Rinse thoroughly | Dry before re-energizing | Straighten fins with comb

9. Anthony's Cool School: https://hvac.anthonyscoolschool.com/

OSHA Safety: https://www.osha.gov/control-hazardous-energy